WASP-135b

R-0191

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_WASP-135_250719` · created 2026-07-16T21:02:11+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2460875.72 +/- 0.014 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.174 +/- 0.05
Transit depth (Rp/Rs)^2	3.04 +/- 1.74 [%]
Orbital Inclination (inc)	85.22 +/- 2.91
Airmass coefficient 1 (a1)	10.59 +/- 2.99
Airmass coefficient 2 (a2)	-2.27 +/- 1.04
Scatter in the residuals of the lightcurve fit is	28.01 %
Best Comparison Star	#3 - [561, 325]
Optimal Aperture	2.41
Optimal Annulus	5.25
Transit Duration (day)	0.081 +/- 0.013

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.174 ± 0.05 vs 0.139 (deviation 0.69σ)
Duration fit vs published	0.081 d vs 0.0687 d (deviation 0.93σ)
Mid-time O–C	-34.8 min
Depth SNR	3.4
Residual scatter	28.01 %

Light curve

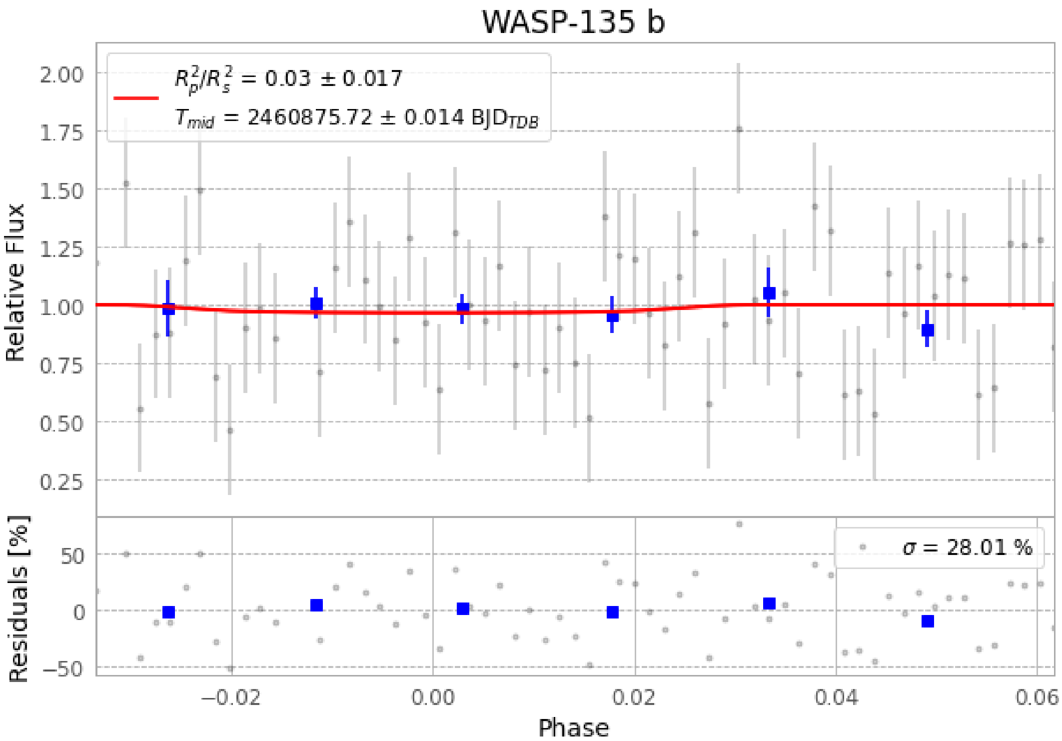

Figure 1 — FinalLightCurve_WASP-135 b_2025-07-18.png (SHA-256 78e12e7c0e5ea349…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [514, 112], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 09c7721baab6c59a…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 378b18d`

Data provenance

65 FITS frames (43 MB), `WASP-135250719040315.FITS` → `WASP-135250719071515.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-135-250719.log (e98b19dd868d…), FinalLightCurve_WASP-135 b_2025-07-18.png (78e12e7c0e5e…), FinalLightCurve_WASP-135 b_2025-07-18.csv (1cb86ce9c5ee…)

Compute resources

Wall time 150.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)